Homoscedasticity vs Heteroscedasticity

Homoscedasticity

- Homoscedasticity means the spread of errors is the same everywhere in the data.
- If we draw dots on a graph, the dots will be spread evenly around the line from the beginning to the end. So, the distance between the dots and the line stays similar.

Heteroscedasticity

- Heteroscedasticity means the spread of errors is not equal across the data.
- On a graph, the dots start close together but become more spread out (or the opposite). So, the distance between the dots and the line changes as values increase.

Purpose

- The purpose of checking this is to make sure our regression model is reliable.
- If the error spread is consistent, the model's predictions and statistical tests become more trustworthy.

Importance

Checking homoscedasticity is important because:

- ✓ It is an assumption of linear regression
- ✓ It helps ensure accurate prediction results
- ✓ It ensures valid hypothesis testing (p-values, confidence intervals)
- ✓ It helps the model behave fairly across all values

If heteroscedasticity exists, the model may give misleading results.

Thumb Rule

When we plot Residuals vs Predicted Values:

- ✓ Random spread with equal width → Homoscedasticity (Good) - because the model errors are stable and reliable.
- ✓ Cone shape / funnel shape / increasing spread → Heteroscedasticity (Problem) - because the error changes across the data and may affect model accuracy.

When Do We Check This?

We check this after building a regression model, usually during:

- ✓ Regression diagnostics
- ✓ Residual analysis
- ✓ Model evaluation